Philip (Phi) Bui

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EDUCATION

Ph.D. in Urban Systems, New York University (NYU), New York, NY Sep 2026 - Present

- Studying honest uncertainty in urban infrastructure sensing.
- Thesis: Quantify, Standardize, Mitigate: Uncertainty in 3D Models of Urban Environments for Autonomous Decision-Making.
- Advised by Dr. Debra Laefer, Urban Modeling Group.

M.S. in Computer Science: Human-Robot Interaction, Tufts University, Medford, MA Sep 2024 - May 2025

- Graduated with a 3.8/4.0 GPA.

B.S. in Computer Science, Worcester Polytechnic Institute (WPI), Worcester, MA Aug 2021 - Dec 2023

- Graduated with a 3.8/4.0 GPA and distinction; Dean's List.

HONORS & AWARDS

Tandon School of Engineering, NYU, New York, NY 2026

- School of Engineering Ph.D. Fellowship, $95,119.

Department of Environmental Protection, NYC, New York, NY 2026

- Cyvl's winning submission to the Environmental Technology Lab (ETL) Challenge, selected to the 2026 cohort.

Jonathan M. Tisch College of Civic Life, Tufts University, Medford, MA 2025

- Inducted into the Honos Civicus Society for civic impact through Untitled LLC.

School of Arts and Sciences, WPI, Worcester, MA 2021 - 2023

- Inducted into Upsilon Pi Epsilon (Computer Science Honor Society).

TECHNICAL SKILLS

Geospatial & sensing: LiDAR point clouds, georeferenced street-level imagery, GIS layers and open municipal data, Google Earth Engine, Qdrant vector search, Model Context Protocol (MCP) servers.
Machine learning & robotics: reinforcement learning (DQN, PPO, multi-agent), LLM fine-tuning and quantized local inference, LLM reward models, Monte Carlo localization, probabilistic roadmaps, HighwayEnv.
Measurement & evaluation: proper scoring rules, calibrated confidence, measurement-error and regression-to-the-mean analysis of survey panels, live-data benchmarks, human-subject (HRI) studies.
Languages: Python, TypeScript/JavaScript, C#, Java, SQL.
Software: React, Node.js, ASP .NET Web API, PostgreSQL, Docker, REST APIs, WebSocket, PyPI packaging, Git, Agile, Shape Up

PUBLICATIONS & PRESENTATIONS

State-Wise Constrained Policy Shaping for Zero-Shot Runtime Behavior Steering, Tufts University 2026

- Second author, with Thomas Howell, Robert McPherson, and Dr. Vasanth Sarathy.
- Peer-reviewed workshop paper. Presented at the AAAI-26 AIGOV workshop in Singapore.
- Developed a post-training supervisor enforcing safety norms on DQN agents in HighwayEnv; reduced collisions by 97% in-distribution and 99% zero-shot.
- Full-paper version, State-Wise Constrained Policy Shaping for Runtime Normative Supervision, under review at the AAAI-27 Main Technical Track (advanced to Phase 2, September 2026).

Scoring Rules Certify Reporting, Not Seeking: A Dilemma for Level-Based Rewards in Sequential, Embodied Perception 2026

- Sole-author manuscript showing that level-based strictly proper scoring rules certify honest reporting but not honest information-seeking, holding for every strictly proper binary rule via Savage's representation.
- Presented as a poster at the IROS-26 Full-Shift Robot Co-Workers workshop (non-archival) in Pittsburgh.

GailBot: A PyPI-Distributed Transcription Pipeline for Conversation-Analytic Research, Tufts University 2026

- Lab demonstration with Dr. Saul Albert, Dr. Umair Muhammad, and Dr. J.P. de Ruiter at the ICCA-26 workshop at University of Alberta: "Using AI within CA Workflows" (facilitated by Dr. Saul Albert and Dr. Lucas Seuren).

RESEARCH EXPERIENCE

Research Intern, AI Futures Collab, WPI, Worcester, MA May 2025 - Sep 2025

- With Dr. Yunus Telliel. Investigated human/AI interfacing and literacy, evaluating phenomenology, affordances, AI interactions, transparency and ethical implications of AI-driven media.

Swarm Robotics AI Research Intern, Air Force Research Laboratory, Eglin Air Force Base, FL May - Aug 2025

- With Dr. Scott Nivison. Designed heterogeneous multi-agent reinforcement-learning algorithms for swarm coordination and decentralized decision-making under communication constraints. (Classified; details omitted.)

Graduate Research Assistant, Human Interaction Lab, Tufts University, Medford, MA Jan - May 2025

- With Dr. Saul Albert, Dr. Umair Muhammad, Dr. J.P. de Ruiter.
- Led the back-end overhaul of GailBot, a PyPI-distributed transcription pipeline for Jeffersonian conversation-analytic notation, built atop off-the-shelf speech-to-text (Deepgram, Watson, Google, OpenAI).
- Engineered REST APIs, designed the PostgreSQL schema, implemented full CRUD in React; authored cross-platform deployment docs and resolved critical Linux/macOS regressions.

PROJECTS

Gestalt, Independent Project Aug 2026 - Present

- Built local search over Markdown notes for Claude Code and Cursor: stella_en_400M_v5 dense vectors in SQLite, BM25 as the lexical fallback, and an optional Qwen3-Reranker-0.6B rerank, with session hooks and skills. Scores nDCG@10 of 0.794 on BEIR SciFact, 0.422 on NFCorpus, and 0.572 on FiQA, reported with 95% bootstrap intervals and paired permutation tests. Released under Apache-2.0 with a Zenodo DOI (10.5281/zenodo.23324594). Code, per-query results, and one-command rerun: github.com/phbui/gestalt-core.

Regression to the Mean in Municipal Pavement Condition Panels: Why Year-over-Year Change Overstates Condition-Dependent Deterioration, Independent Research Aug 2026

- Sole-author manuscript quantifying measurement error and regression to the mean in municipal pavement-condition survey panels from multi-epoch public archives (New York City, San Francisco). In the New York City panel, decomposes rating variance into rater error (~15%) and non-persistent occasion effects (~38–56%). Manuscript in preparation.

Do Agents Disclose What the Data Cannot Support? A Live-Data Benchmark for Abstention, Measurement-Uncertainty Caveats, and Missingness Disclosure over Municipal Open Data, Independent Research **Aug - Oct 2026**

- Sole-author manuscript, under review at ACM FAccT 2027, presenting a 596-question live-data benchmark over a municipal open-data API, run on ten model settings from six families through one protocol, with the evaluation environment manipulated as a factor.
- Every setting carries a documented reliability caveat into 100% of answers, yet discloses a discoverable data defect in 0% to 33% of eligible answers; handed the defect's null count in a readable schema, six of ten settings disclose in 44% to 96% and four do not move, and one added paragraph that tells the agent to check the null share raises disclosure to 74% to 99% in all five settings that ran it. Stated confidence carries little information about correctness (AUROC at or below 0.5 in 17 of 35 run-settings).
- Benchmark, dataset, and 44,541 episode records publicly released (v1.3.0, CC-BY-4.0 data, MIT code). Code and data: github.com/phbui/civic-honesty-benchmark.

LLM Enhanced Captioning for Neurodivergent Users, Tufts University Jan - May 2025

- Co-developed a real-time Chromium extension integrating a quantized Mistral-7B model over WebSocket; enriched captions with sentiment and intent, achieved <200 ms latency & >95% word accuracy; two pilots (n=8, n=3) reported 4.5/5 clarity. Code: github.com/phbui/real-time-captioning-extension.

Safe Reinforcement Learning for Autonomous Driving via Metrics-Driven Normative Supervision, Tufts University Jan - May 2025

- Implemented a metrics-driven normative supervisor as a non-learned policy layer over a DQN agent in HighwayEnv, with quantitative norms for speeding, tailgating, sudden braking, and lane changes; up to 89% fewer collisions and strong zero-shot generalization across three custom test environments.
- Improved the supervisor into the peer-reviewed AAAI-26 AIGOV workshop paper (see Publications & Presentations).

LLMs Can Be Judgy Too: Fine-Tuning LLMs for Character-Driven RL, Tufts University Dec 2024 - May 2025

- Built a PPO pipeline using a fine-tuned LLM as a reward model for facial-AU generation; across 30 scenarios, LLM-only achieved 3.12/5 mean accuracy, the hybrid approach matched at 3.07/5 with 32% less variance, and halved manual labeling effort.

Computer Vision-Guided Sphero Swarm Navigation Framework, Tufts University Sep - Dec 2024

- Integrated overhead-camera Monte Carlo localization (MCL), probabilistic roadmap (PRM) path planning, and a Bluetooth Low Energy (BLE) microservice layer to enable scalable, real-time collision-free coordination of Sphero Bolt swarms. Code: github.com/phbui/sphero-swarm.

Evaluating Preferences in Robot Control Systems, Tufts University Sep - Dec 2024

- Ran a human-robot interaction (HRI) study (n=11) comparing manual stick control vs. autonomous MCL/PRM navigation on obstacle courses; manual was 30% faster with slightly fewer collisions and preferred by 64% of participants.

Haven, Tufts Producthon Sprint, Tufts University Feb 2024

- Built a map that locates harm-reduction resources and heat-maps anonymously reported incidents so college students can warn their community.
- Won 1st place in the Health/Wellness category.

AI Platform for Low-Income Housing Development for City of Worcester, MA, WPI Major Qualifying Project (MQP) Aug - Dec 2023

- With Dr. Daniel Treku, Dr. Matthew Ahrens, Dr. Wilson Wong, Prof. Robert Sarnie, Prof. John Galante, Prof. Suzanne LePage, and Prof. Sara Saberi. Built all of the platform's software. Partnered with Worcester's Chief Development Officer to build an AI-assisted land-acquisition and financing pipeline. Integrated GIS layers and open municipal data into a PostgreSQL/Node.js backend and React dashboard.
- Surfaced parcel-level transparency for civic stakeholders. Published records onto blockchain for transparency and accountability. A homeless shelter and rehabilitation clinic broke ground in record time as a result; the City of Worcester now owns and operates the platform in its low-income housing development pipeline. Code: github.com/FinTech-MQP/Worcester-PermitPro.

Nitrogen Cycle Public Outreach Augmented Reality Game, WPI Interactive Qualifying Project (IQP) Oct 2022 - Mar 2023

- Product manager and front-end developer for an augmented-reality incremental game that teaches the nitrogen cycle, built with the WPI STEM Education Center for the Boys & Girls Club of Fitchburg and Leominster; designed the interface and mechanics and delivered them through storyboards and testing rounds. Code: github.com/IQP-NCPOGD/nitrocycle.

Medical Service Request System for Brigham & Women's Hospital, WPI Software Engineering (CS 3733) Mar - Apr 2022

- Full-stack engineer on an eleven-person Agile team building a Java equipment-tracking and service-request application with a hospital dashboard and floor-map editor; led the interactive visual simulation subgroup and designed the feature subsystems through UML use-case, class, sequence, and activity diagrams.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Cyvl, Somerville, MA Aug 2025 - Aug 2026

- Designed the computer-vision ingestion and data-fusion pipeline that maps NYC's underground infrastructure from non-standard municipal engineering documents and diagrams into

geometrically-accurate geospatial files; this became the winning submission to NYC DEP's ETL Challenge, earning selection to the 2026 ETL cohort.
- Built a visual embedding pipeline indexing 500k+ high-resolution street-level images per city into a Qdrant vector store, enabling sub-second semantic retrieval over urban infrastructure imagery.
- Extended production AI pipelines processing petabyte-scale, georeferenced imagery and LiDAR point clouds to identify and analyze pavement distress, signs, markings, and above-ground assets across road networks.
- Shipped dozens of features adopted by hundreds of executive-level stakeholders across municipal public-works and transportation departments.
- Partnered with City of Boston leadership to build and maintain Model Context Protocol (MCP) servers, giving non-technical civic staff natural-language access to infrastructure datasets and narrowing the data-access gap between city hall and the agencies that serve residents.
- Investigated calibrated-confidence methods for infrastructure change-detection as an internal feature; internal findings on measurement inconsistency in repeat surveys and on reward design for honest reporting seeded an independent academic research program (see Research Experience).

Software Engineer, enLabel, Boston, MA Jan 2024 - Jan 2025
- Refactored hundreds of defects in legacy C#/ASP .NET code; led React/TypeScript UI redesign, improving workflows and user satisfaction.
- Architected a modular React front end with optimized ASP .NET Web API integration; authored end-to-end regression-testing suite to automate API validation.
- Contributed to technical planning and architecture meetings with cross-functional stakeholders.
- Led design specification, timeline management, and strategic coordination of engineering deliverables.

Website Developer, ProsperOn, Boston, MA Jun 2023 - Feb 2024
- Built accessible React/TypeScript financial-simulation pages from Figma; refactored Node.js RESTful APIs to modernize legacy code.
- Piloted platform integrations with MassMutual, Mass Fintech Hub, and Boston University; featured in Fintech Sandbox as a leading financial-wellness startup.

Operations Intern, Amazon, Salem, OR Jun - Aug 2023
- Developed a Node.js mapping tool to track >7,000 items per warehouse; integrated into onboarding training, cutting per-employee costs by 95%.

COMMUNITY SERVICE

Community Team Coordinator, Cyvl, Somerville, MA Sep 2025 - Aug 2026
- Instructed the "Infrastructure Intelligence" sessions for the Public Innovation Institute × Cyvl Summer Innovation Program, teaching students to work with municipal infrastructure data and tooling.
- Developed system of community outreach to institutions, university organizations, and data communities such as: Yale (PITSCY), NYU (CUSP), Tufts (JumboHacks), BU (Spark), NEU (SUN Lab), WPI (UPE), Olin, BetaNYC (NYC School of Data).
- Collaborated with university organizations to bring lectures, workshops, sponsorships, hackathons, and research opportunities to students.

Co-Founder & Chief Logistics Officer, Untitled LLC, Boston, MA Dec 2023 - May 2026
- Launched an underground arts collective securing venues, sponsorships, and studio space; produced sold-out fashion shows and performances for thousands of attendees and millions of social-media views.
- Recruited/managed cross-functional teams, handled legal/financial strategy, coordinated security & harm-reduction, and led fundraising for underrepresented artists.

Front End Developer & Educator, Freelance, Boston, MA Feb 2023 - Present
- Collaborated with local artists to translate creative visions into responsive digital experiences, and educated them on front-end fundamentals.

Community Events Coordinator, WPI Powerlifting, Worcester, MA Jan 2023 - Dec 2023

- Forged partnerships with collegiate and professional powerlifting organizations, including a collaborative lift with BU Barbell; designed inclusive skill-building workshops such as the Empowerment Workshop, boosting recruitment and diversity.

Lifeguard, Swim & Gym Instructor, YWCA Central Massachusetts, Worcester, MA Aug 2021 - Jun 2022

- Delivered swim & fitness instruction to 6–65 year-olds, including students with disabilities; achieved 100% course completion and performed three lifesaving interventions.